

RITI SHARMA

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PhD candidate in Mechanical Engineering and Applied Mechanics at the University of Pennsylvania working at the interface of mechanical engineering, materials science, and biology. I study bone as a complex hierarchical system to understand how structure, mechanics, and time-dependent behavior govern damage resistance. My work combines experimental design, in-situ imaging, and data analysis to uncover links between material properties and biological function. Particularly interested in viscoelastic and soft biomaterial systems (including polymer-like and hydrated biomaterials) and their role in cellular and tissue-level responses.

Interests: Biomaterials, viscoelasticity, cell-material interactions, fatigue and fracture, bio-inspired material design.

Education

Doctor of Philosophy in Mechanical Engineering and Applied Mechanics (ongoing)

Sept 2022–Present

University of Pennsylvania, Philadelphia, USA

Research focus: Nanoscale mechanical behavior, viscoelasticity, and fatigue of biomaterials (bone) in 3D

Master of Engineering (research) in Bioengineering

Jul 2020–Jul 2021

Auckland Bioengineering Institute, University of Auckland (UoA), New Zealand

First-class honors, [Abstract](#) and [Python Implementation](#) available online for research use

Bachelor of Technology (Eng.) in Mechanical Engineering

Aug 2016–May 2020

Manipal Institute of Technology, India

Research Experience

Researcher, PhD in Mechanical Engineering and Applied Mechanics

Sep 2022–Present

University of Pennsylvania, Philadelphia, USA

- Designed and executed iterative, hypothesis-driven experiments to characterize structure–property relationships in biological materials (bone), from setup to analysis and interpretation, adapting protocols based on emerging data and experimental constraints.
- Investigated viscoelastic and time-dependent deformation mechanisms in biological tissues, interpreting them as soft, composite biomaterials and drawing parallels to polymeric and viscoelastic systems relevant to engineered biomaterials.
- Developed Python and MATLAB pipelines for 3D image processing, quantitative characterization, and interpretation of large experimental datasets.
- Collaborated with bioengineering researchers to relate material behavior to cellular and microstructural context, connecting mechanics with biological function.

Research Assistant

Aug 2021–Aug 2022

Auckland Bioengineering Institute, University of Auckland, New Zealand

- Conducted functional experiments, structural imaging, and computational analysis for studies of cardiomyocyte (heart cells) energy metabolism.
- Worked at the interface of mechanics and biology to analyze functional behavior in cellular systems.

Research Intern

Jan 2020–Jul 2020

Auckland Bioengineering Institute, University of Auckland, New Zealand

- Applied Amira, ImageJ, MATLAB for 3D construction and segmentation of human atria MRI images
- Conducted cardiac structural analysis, modeling, atrial fibrillation simulations, and mapping electrophysiology using mathematical models

Technical Skills

Mechanical Testing: Fatigue, fracture, cyclic loading, in-situ characterization, failure analysis, viscoelastic behavior, stiffness-compliance analysis, micro-and nano-mechanics.

Biomaterials and Materials Characterization: Electron microscopy (SEM, TEM), Optical microscopy, synchrotron X-ray tomography, radiography experiments, Raman spectroscopy.

Data Analysis and Computational Tools: MATLAB, Python, ImageJ, Amira, Dragonfly, high-resolution segmentation, volumetric reconstruction, quantitative image analysis, data visualization, simulation.

Additional Experience

Research Intern

Jun 2018-Jul 2018

Indian Institute of Technology (BHU), India

- Performed Computational Fluid Dynamics simulations of diffusion in 2D plates.

Teaching Assistant, University of Pennsylvania, USA

Spring '24, Fall '24, Spring '25

- Led recitations and designed assignments for Dynamic Systems and Control, Statics and Strength of Materials

Mechanical Engineering Intern, Honda Cars India Ltd, Greater Noida, India

May 2019-Jun 2019

- Conducted Process Studies to reduce MUDA (waste) in the Assembly Frame department.

- Applied analytical problem-solving to improve manufacturing efficiency in a high-volume production environment.

Patent Intern, Davies Collison Cave, New Zealand

Sept 2021 - Oct 2021

- Reviewed and drafted mechanical engineering patents, conducting infringement-related searches, and analyzing examiner responses.

Mentorship and outreach, University of Pennsylvania, USA

Apr 2023-Aug 2024, Spring '26

- Mentored one high school student, three undergraduate students.

- Partnered with a high school teacher to establish a curriculum to bridge the gap between school students and research labs.

Vice President, Mechanical Engineering Graduate Association (MEGA), University of Pennsylvania, USA

Oct 2023-Oct 2024

- Organized social events, communicated student needs, and fostered a sense of community among graduate students

Business and Strategy Intern, Mercury Energy and University of Auckland, New Zealand

Dec 2020

- Proposed a business model to facilitate electric vehicle uptake among businesses in New Zealand and Australia

Selected Presentations and Publications

1. **R. Sharma**, S. Ching, L. Capaldi, K. Chen, X. Xiao, O. A. Tertuliano. Bone Resists Fatigue through Crack Deceleration at the Fibril Scale. *Science Advances*, Manuscript in Revision, 2026. bioRxiv doi: <https://doi.org/10.1101/2025.09.30.679500>
2. **R. Sharma**, Bone Resists Fatigue through Crack Deceleration at the Fibril Scale. **Invited talk**, Environmental and Biological Mechanics Seminar Series, University of Pennsylvania, 2026.
3. **R. Sharma**, Bone Resists Fatigue through Crack Deceleration at the Fibril Scale. Poster and Talk, Society of Engineering Science Technical Meeting, 2025. **Travel Award**.
4. **R. Sharma**, Resolving deformations of mineralized collagen fibrils in 3D. Poster, Summer Biomechanics, Bioengineering, and Biotransport Conference, 2024.
5. **R.Sharma**, Resolving deformations of mineralized collagen fibrils in 3D. Poster, Penn Center for Musculoskeletal Disorder Symposium, 2023. **Poster Award**.
6. **R.Sharma**, Resolving deformations of mineralized collagen fibrils in 3D. Poster and Presentation, Society of Engineering Science Technical Meeting, 2023. **Poster Award**.